IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-351

1. Topic (12 words max)

AI-Powered Kitchen Ecosystem for Food Waste Reduction and Nutrition Optimization

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	-
Year	-

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Food waste represents one of the most pressing global challenges, with approximately 40% of food produced worldwide being wasted annually, contributing to environmental degradation, economic losses, and social inequality. Simultaneously, many individuals struggle with meal planning, nutrition tracking, and maintaining motivation for healthy cooking practices, leading to poor dietary choices and increased reliance on processed foods.

Current food technology solutions offer fragmented approaches to these interconnected problems. Recipe generation applications focus solely on meal suggestions without considering ingredient availability or freshness. Nutrition tracking apps provide calorie counting but lack real-time behavioral guidance. Grocery planning tools generate basic shopping lists without optimizing for waste reduction or household consumption patterns. Most critically, no existing system addresses the complete food lifecycle from planning to waste management in an integrated manner.

This research addresses the significant gap in comprehensive smart kitchen technology by proposing an AI-powered assistant that combines multiple domains: computer vision for ingredient recognition, natural language processing for recipe adaptation, behavioral AI for personalized guidance, and optimization algorithms for waste reduction. The system uniquely integrates pantry intelligence with cultural recipe adaptation, dynamic nutritional nudging, predictive grocery planning, and social cooking support.

The problem is particularly relevant in developing countries where food security concerns intersect with rapid urbanization and changing dietary patterns. Traditional cooking knowledge is being lost while processed food consumption increases, creating both nutritional and environmental challenges that require technological intervention.

References:

Gustavsson, J., Cederberg, C., Sonesson, U., van Otterdijk, R., & Meybeck, A. (2011). Global food losses and food waste: extent, causes and prevention. FAO.

Chi, P., & Demiris, G. (2007). Enabling nutrition-aware cooking in a smart kitchen. Proceedings of the SIGCHI Conference on Human Factors in Computing Systems.

Zhang, Y., & Wang, S. (2023). FoodLMM: A versatile food assistant using large multi-modal model. arXiv preprint arXiv:2312.14991.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

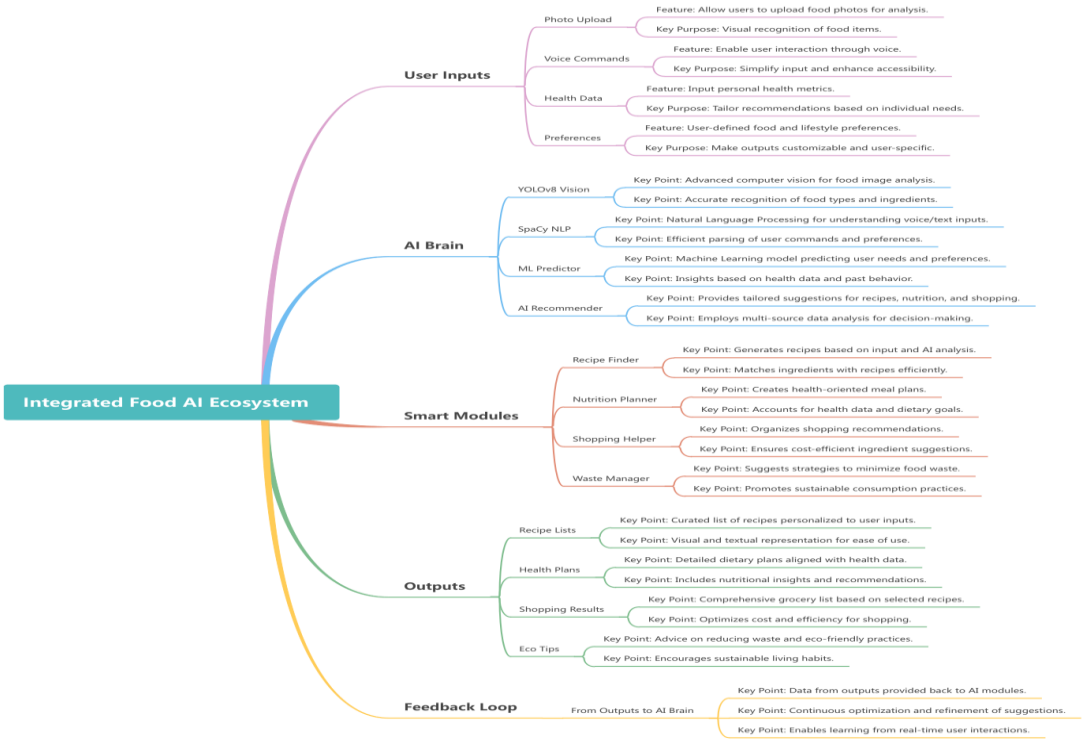
The proposed Spontaneous Cooking Assistant leverages advanced AI technologies to transform random ingredient photography into actionable recipe suggestions. The system employs YOLOv8-based multi-object detection to simultaneously identify diverse food items including fresh produce, packaged goods, and condiments within a single photograph.

The core innovation lies in contextual recipe matching algorithms that understand ingredient relationships and cooking possibilities. When users upload photos of available ingredients, the system processes the image through computer vision models, identifies individual items with confidence scoring, and queries comprehensive recipe databases to suggest achievable meals based on ingredient overlap percentages.

System Architecture: User captures photo of available ingredients → AI processes image using multi-object detection → System identifies individual food items with confidence scores → Recipe matching algorithm evaluates ingredient combinations → System presents ranked recipe suggestions with missing ingredient notifications → User feedback improves future recognition accuracy.

The solution incorporates intelligent error correction through user feedback loops, enabling the system to learn from misidentifications and improve recognition accuracy over time. This creates a self-improving cooking assistant that becomes more reliable through real-world usage.

The integrated approach addresses the practical cooking scenario of "what can I make with these random ingredients?" while supporting sustainable cooking practices by maximizing ingredient utilization and reducing food waste through creative recipe suggestions.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

The development requires specialized expertise in advanced computer vision for multi-object food detection, including proficiency in YOLOv8 architecture, object detection algorithms, and image preprocessing techniques for handling diverse kitchen photography conditions. Deep learning expertise is essential for fine-tuning pre-trained models on food-specific datasets and implementing confidence scoring mechanisms for recognition accuracy.

Algorithm development knowledge supports the creation of contextual recipe matching systems that evaluate ingredient combinations and cooking relationships. This includes expertise in similarity scoring, database optimization for real-time querying, and ranking algorithms that prioritize achievable recipes based on available ingredients.

Machine learning specialization in feedback loop implementation and progressive learning systems enables the development of self-improving recognition accuracy through user corrections and behavioral adaptation.

Data requirements include comprehensive food recognition datasets such as Food-101 for basic classification, Recipe1M+ for recipe-ingredient mapping, FoodX-251 for fine-grained food categorization, and Grocery Store Dataset for packaged goods recognition. Custom data collection will supplement existing datasets with diverse ingredient combination photography under various lighting conditions and camera angles.

Technical infrastructure expertise encompasses mobile application development for intuitive photo capture interfaces, cloud computing for scalable AI model deployment, API development for recipe database integration, and database management for handling large-scale ingredient-recipe mappings while maintaining rapid query response times.

User experience design knowledge supports the creation of error correction workflows, confidence visualization, and intuitive recipe presentation interfaces that facilitate seamless interaction between users and the AI recognition system.

8. Objectives and Novelty

Main Objective

To develop an intelligent AI-powered smart kitchen assistant platform that integrates computer vision, natural language processing, and behavioral AI to reduce household food waste by up to 30%, promote healthier eating habits through personalized nutritional guidance, and enhance social cooking experiences by providing comprehensive support from meal planning to waste management, thereby creating a sustainable and user-friendly ecosystem that addresses the complete food lifecycle while adapting to individual household needs and cultural preferences.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Perera U.M.K. IT22131942	To develop an intelligent spontaneous cooking assistant that recognizes multiple food items from a single photograph and suggests recipes based on available ingredients, supporting real-world impulse cooking scenarios with self-improving accuracy through user feedback	• Implement YOLOv8-based multi-object detection to simultaneously identify diverse food items (fresh produce, packaged goods, condiments) in a single image • Develop an image preprocessing pipeline to handle various lighting conditions, camera angles, and kitchen environments • Create a confidence scoring system to assess recognition accuracy for each detected item. • Design an algorithm to match detected ingredients with recipe	Spontaneous Cooking Support Innovation: Unlike existing meal planning systems that assume predetermined recipes, this module addresses the common real-world scenario of "I have random ingredients, what can I make?" This impulse cooking assistance represents a significant gap in current food technology applications. Multi-Modal Recognition with Context: The system goes beyond single-item food recognition to understand ingredient relationships and cooking possibilities from arbitrary combinations. This contextual understanding of how diverse

		database using ingredient overlap scoring • Implement recipe ranking system based on ingredient availability percentage and cooking complexity • Develop missing ingredient identification and substitution suggestion features • Create user feedback interface for correcting misidentified food items • Implement progressive learning system that improves recognition accuracy over time • Develop similarity-based suggestion system for ambiguous food items • Build recipe suggestion engine that prioritizes recipes achievable with available ingredients • Create difficulty-based recipe categorization (easy/medium/complex) based on cooking skills required • Implement alternative recipe suggestions when primary ingredients are missing	ingredients work together is not addressed in current food recognition systems. Self-Improving Recognition System: The integration of user feedback loops with progressive learning creates a system that becomes more accurate over time through real-world usage, addressing the common problem of AI food recognition errors in practical applications.
Shahmi M.T.M IT22083982	The system provides AI-driven, personalized nutritional	• Collect user data like age, weight, allergies, sleep, and diet	Unlike traditional nutrition systems that focus on static meal plans or reactive calorie tracking, this research introduces

	recommendations based on a user's lifestyle, biometrics, and preferences, proactively predicts micro-nutrient deficiencies from intake patterns and cooking behavior, dynamically optimizes meals based on work schedules and sleep cycles, and supports mental health through stress-responsive, mood-balancing meal planning.	preferences for personalized nutrition. • Nutrient intake is tracked by logging meals or scanning recipes for detailed analysis. • Machine learning model predicts micro-nutrient deficiencies using food intake and biometric data. • The circadian nutrition engine suggests meals based on the body's natural rhythms. • Stress-responsive engine detects emotions from voice input to adjust meal recommendations. • The system adapts meal suggestions according to user goals and progress. • Feedback module provides progress tracking and motivational nudges for habit building. • Module that alerts users about potential negative interactions between their medications and certain foods • Predict possible allergic reactions or food intolerances in advance by analyzing historical intake patterns and reported symptoms.	a novel, AI-powered framework that delivers proactive and personalized nutritional guidance. The system integrates predictive micro-nutrient deficiency detection using advanced AI modeling, circadian-aware meal optimization aligned with biological rhythms and lifestyle patterns, and emotion-driven nutrition that dynamically adapts to stress and mood through voice and user input. It also incorporates real-time medication-food interaction alerts and predictive allergy and intolerance risk assessments, enhancing user safety and personalization. By leveraging multi-modal data, including biometric signals, behavioral logs, and contextual factors such as seasonal and cultural influences. The framework provides deeply contextualized, adaptive recommendations. Additionally, behavior-aware adaptive feedback loops support long-term dietary habit formation and transformation. This multi-dimensional approach addresses significant gaps in existing nutrition technologies by combining food science, chronobiology, emotional intelligence, clinical safety, and behavior modeling into a unified, intelligent solution.

D.H Jayasundara IT22117946	The part aims to help users find products that match their budget and preferences. It allows voice-based searches for hands-free experience. AI and natural language processing (NLP) are used to make searches smarter. The system pulls product data from websites like eBay and Walmart for comparison. It also supports price displays in different currencies. Users can save, view, update, or delete their search history. Finally, it provides an easy-to-use interface for a smooth experience.	• The project involves building a Flask backend to manage APIs for recommendations and search history. It uses Selenium and BeautifulSoup to scrape product data from eBay and Walmart. • SpaCy extracts keywords from user queries, and OpenAI analyzes reviews. • Voice recognition is added using Python and React. Search history is stored and managed with options to view, update, or delete. • Currency conversion is implemented with forex-python. A React frontend is created with search, filters, sorting, and a chat feature.	This project stands out by offering voice-controlled shopping, which is rare in e-commerce apps. It uses AI and NLP for smarter product suggestions and review analysis. It compares prices across multiple websites in real time. The ability to manage search history adds personalization. Currency conversion makes it usable worldwide. A chat assistant that understands context acts like a real shopping helper.

Muraleswaran D IT22339010	The system uses AI to detect household food scraps through image recognition and provides smart, actionable suggestions for composting or reusing those scraps. It integrates community-based compost hub mapping, tracks composting habits over time, and offers sustainable alternatives to minimize food waste. By turning discarded scraps into a usable resource or eco-friendly action, the system promotes environmental responsibility at the household level.	• Users upload images of leftover or discarded food scraps (e.g., vegetable peels, fruit rinds, spoiled items). • Image classification model is used to detect and categorize types of food waste. • A rule-based engine matches each waste type to composting advice (e.g., “Add to dry compost pile,” “Avoid citrus in worm bins”) or creative reuse suggestions (e.g., “Use banana peel for fertilizer”). • Integrates with a map service (e.g., Google Maps API) to show nearby composting hubs, donation centers, or collection points. • System tracks user actions and visualizes composting impact over time (e.g., estimated waste diverted from landfills). • Includes a simple reward or badge system to motivate continued sustainable behavior. • Offers educational snippets, tips, or facts about reducing food waste and composting benefits.	Unlike most existing food-related AI systems that focus on meal planning, nutrition, or grocery management, this module takes a sustainability-first approach by addressing what happens after the meal — specifically, how to deal with food scraps and spoilage. It introduces a unique AI system that uses image recognition to detect household food waste and provide smart suggestions for composting or reuse. A built-in rule engine recommends the best composting method based on the type of waste and environmental guidelines. It also connects users with nearby compost hubs or drop-off points, encouraging community-based waste solutions. The system tracks user behavior over time, estimates environmental impact (like waste reduced), and uses gamified features such as badges to promote consistent, eco-friendly habits. By shifting focus from food consumption to responsible disposal, this module fills a significant gap in current food tech solutions.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Perera U.M.K. IT22131942	**Spontaneous Cooking Assistant - Multi-Item Food Recognition and Recipe Discovery System** **Core Innovation:** This module addresses the practical cooking scenario where individuals have random ingredients and need immediate recipe suggestions, representing a significant advancement over traditional meal planning approaches that assume predetermined recipes or shopping lists. **Advanced Computer Vision Implementation:** The system leverages state-of-the-artYOLOv8 object detection to simultaneously identify multiple diverse food items within a single photograph. This requires sophisticated expertise in deep learning architectures, particularly in handling complex scenes with varying object sizes, lighting conditions, and occlusion patterns. The implementation involves fine-tuning pre-trained models on food-specific datasets including Food-101, Recipe1M+, and custom-collected grocery store imagery. **Multi-Object Detection Specialization:** Unlike simple single-item classification, this module handless simultaneous recognition of fresh produce, packaged goods, condiments, and cooking ingredients in realistic kitchen environments. This requires advanced understanding of object detection algorithms, non-maximum suppression techniques, and confidence threshold optimization to ensure accurate identification across diverse food categories. **Contextual Recipe Matching Algorithms:** The system implements sophisticated ingredient-to-recipe mapping algorithms that understand cooking relationships between detected items. This involves developing scoring mechanisms that evaluate ingredient overlap percentages, cooking compatibility, and recipe complexity. The algorithm considers partial ingredient matches and suggests creative cooking possibilities that maximize the use of available ingredients. **Machine Learning and Adaptive Systems:** The module incorporates progressive learning capabilities through user feedback integration, demonstrating expertise in online learning and model adaptation. The system tracks user corrections, builds confidence in recognition patterns, and continuously improves accuracy through real-world usage data. This requires understanding of feedback loops, model updating strategies, and maintaining system performance while learning. **Database Design and Information Retrieval:** The system manages large-scale recipe databases with efficient querying mechanisms for real-time ingredient matching. This involves expertise in database optimization, indexing strategies for ingredient-based searches, and scalable data structures that support rapid recipe retrieval based on available ingredients. **User Experience and Interface Design:** The module creates intuitive interfaces for photo capture, ingredient confirmation, and recipe presentation. This requires understanding of mobile application development, responsive design principles, and user interaction patterns specific to cooking applications. The interface handles error

	correction workflows and provides clear visual feedback for recognition confidence levels.
Shahmi M.T.M IT22083982	**Nutritional Guidance – Predictive Health Intervention System** This module transforms basic dietary tracking into a proactive, intelligent nutritional support system by integrating AI technologies to personalize food recommendations, predict deficiencies, and respond to emotional and physiological cues. **Predictive Health AI Engine:** Leverages machine learning models to analyze dietary patterns, ingredient usage, and user lifestyle data to forecast potential micro-nutrient deficiencies 2–3 weeks in advance**,** moving beyond reactive food tracking toward preventative health intervention. **Chrono Nutrition Optimization:** Implements circadian-aware scheduling algorithms that tailor food suggestions based on the user's biological clock, work schedule, and sleep pattern. **Emotion Responsive Meal Planning:** Applies stress detection (from voice input or manual logs) to dynamically recommend foods that help regulate cortisol, enhance mood, and support mental health. This showcases AI's role in holistic wellness**,** combining psychology with nutrition. **Personalized Nutrition Recommendation Engine:** Uses user profiles (age, weight, allergies, goals) along with real-time food logging to generate contextual recommendations (e.g., high-protein vegetarian lunch), demonstrating strong natural language understanding and adaptive decision-making**.** **Medication-Food Interaction Alert:** This module checks user medications against consumed foods to warn about harmful interactions, ensuring safe and effective nutrition management. **Predictive Allergy and Intolerance Alert:** This system analyzes past meals and symptoms to predict potential allergies or intolerances, helping users avoid foods that may cause adverse reactions. **Interactive Feedback Loop:** Provides adaptive guidance and motivational nudges through a feedback loop based on habit strength, progress markers, and deviation trends, building a dynamic user engagement model that supports long-term behavior change. **Backend AI System Integration:** Unifies various AI subsystems (emotion analysis, circadian nutrition, nutrient analysis, behavior prediction) into a full-stack AI system, showcasing strong backend engineering and real-world AI solution design. **User Interface and Dashboard:** presents a clear, responsive dashboard for tracking nutrient intake, upcoming risks, and daily meal suggestions, highlighting practical AI deployment with intuitive UX design **Data Handling and Scalability:** Implements structured data management for nutrient profiles, ingredient mappings, and user records, ensuring modular system scalability and performance for varied user bases and data volumes.
D.H Jayasundara IT22117946	**AI Shopping Agent – Intelligent E-Commerce Recommendation System** This module transforms online shopping into a smart, personalized experience by integrating AI, natural language processing (NLP), and web scraping to deliver tailored

	product recommendations, voice-based searches, and dynamic price comparisons across multiple platforms. **Product Recommendation Engine**: Uses AI and machine learning to analyze user preferences, budget, and product type, providing personalized product suggestions from eBay and Walmart. It ensures accurate, relevant recommendations by processing real-time data, moving beyond basic search to intelligent shopping assistance. **Voice-Activated Search**: Implements speech recognition using Python's speech_recognition and React's Web Speech API, allowing users to search for products hands-free. This innovative feature enhances accessibility and aligns with modern AI-driven user interaction. **NLP-Powered Query Processing**: Leverages SpaCy to extract keywords from user inputs, refining search accuracy. It understands user intent (e.g., "gaming laptop") and delivers context-aware results, showcasing advanced NLP capabilities in e-commerce. **Multi-Platform Web Scraping**: Employs Selenium and BeautifulSoup to fetch real-time product data (name, price, URL) from dynamic websites like eBay and Walmart. This enables accurate price comparisons, demonstrating robust data extraction for e-commerce applications. **Currency Conversion Module**: Integrates forex-python to convert product prices into user-selected currencies (e.g., USD, EUR, INR). This feature supports global accessibility, making the system practical for diverse users. **Search History Management**: Stores user search queries in a database or local storage, allowing users to view, update, or delete their history. This adds personalization and improves user engagement, aligning with modern e-commerce trends. **Context-Aware Chat Assistant**: Provides a conversational interface in React that responds to user queries (e.g., budget or product preferences) with context-aware replies. This mimics a real shopping assistant, highlighting AI's role in interactive user support. **Backend AI and API Integration**: Combines Flask APIs, AI (OpenAI for review analysis), and scraping tools into a unified backend system. This showcases strong full-stack engineering and real-world AI deployment for e-commerce solutions. **User Interface and Dashboard**: Delivers a responsive React frontend with search forms, filters, sorting options, and a chat interface. It presents product recommendations and history clearly, emphasizing intuitive UX design and practical AI integration. **Data Handling and Scalability**: Manages product data, user preferences, and search history with structured storage, ensuring the system scales efficiently for growing user bases and large datasets, demonstrating robust web development for e-commerce.

Muraleswaran D IT22339010	**Food Waste Redemption – AI Composting Companion** This module delivers an AI-powered system that identifies household food scraps via images and offers intelligent composting or reuse advice. It promotes sustainable waste management by integrating local compost hub data, tracking eco-friendly behavior, and motivating users through gamification. **Computer Vision & Image Processing:** Uses Python libraries (e.g., TensorFlow, PyTorch, OpenCV) to develop and deploy an image classification model that detects and categorizes food scraps from user-uploaded photos, demonstrating expertise in AI, image processing, and machine learning. **Rule-Based AI System Development:** Implements a Python-based rule engine to recommend composting methods or reuse strategies depending on scrap type and composting method suitability, showcasing AI logic and decision system skills. **Backend AI Development:** Builds the backend server using Python frameworks (Flask or FastAPI) to handle image uploads, invoke AI models, process composting rules, manage user sessions and data analytics, and serve RESTful APIs for frontend communication, reflecting backend architecture and integration capabilities. **AI Integration with Local Ecosystem Data:** Integrates geolocation and mapping APIs to identify and display nearby composting hubs or food scrap drop-off centers, using Python for external API communication and data processing. **User Behavior Tracking & Analytics:** Tracks composting activity over time, estimates environmental impact (e.g., waste diverted), and visualizes progress using Python-based data analysis and reporting tools. **Frontend Development:** Uses React to create a responsive and interactive user interface for uploading images, viewing composting advice, tracking eco-behavior, and engaging with gamification elements like badges and milestones. React communicates with the Python backend via REST APIs to deliver a seamless user experience. **Sustainable AI Application Development:** Combines AI, backend, and frontend technologies to develop a real-world sustainable solution, demonstrating applied AI skills alongside software engineering principles tailored for environmental impact.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Mrs.	Lokesha	Weerasinghe	
Co-Supervisor	Mrs.	Chathurya	Kumarapperuma	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

<u>This part is to be filled by the Topic Screening Staff members.</u>

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	✓	No	

b) Does the proposed topic exhibit novelty?

Yes	✓	No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes	✓	No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	✓	No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

Recommended to continue the research.

IT4010 Research Project – 2025 July Topic Assessment Form

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.